How Robust Are Foundation-Model Camouflage Detectors? Corruptions, Input Resolution and a Laptop-Scale Adapter

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***Abstract*—Camouflaged object detection (COD) is deployed where cameras blur, add noise and compress, yet COD models are evaluated on clean images. Detectors built on the Segment Anything Model (SAM) now lead clean benchmarks, but their robustness has not been measured. We benchmark three specialist COD models, two SAM-based models and a zero-shot SAM probe under eight common corruptions at five severities on four datasets, with a held-out protocol that separates corruptions used in training from unseen ones. The SAM2-based model is the most accurate and the most robust; the largest model, SAM-Adapter, falls behind it under corruption. Noise, blur, JPEG and rain cause almost all of the damage, and models that read 1024-pixel inputs lose the most under them. We also train a lightweight adapter on a frozen MobileSAM encoder (3.71 M trainable parameters) on a 4 GB laptop GPU. On COD10K it outperforms all specialist models under corruption (mean S_α 0.812 vs. 0.795 for SINet-V2). Trained at 512 pixels it keeps the same accuracy under local corruptions, reaches the retention of SINet-V2 and runs in 23.1 ms per image. Corruption training transfers modestly to unseen corruptions. Code, corruption seeds and per-image scores are released.**

***Index Terms*—camouflaged object detection, robustness, image corruptions, Segment Anything, parameter-efficient adaptation**

I. Introduction

Camouflaged object detection (COD) segments objects that blend into their background [1], [2]. Its applications, such as wildlife monitoring with camera traps and drones, search and rescue, and visual inspection, rarely provide clean images: moving cameras blur, small sensors add noise, links compress, and weather degrades the scene. Camouflage cues are subtle, a faint boundary or a small break in texture, and these are exactly what such degradations remove.

Detectors built on the Segment Anything Model (SAM) [3] now report the best clean COD results [4], [5]. Their behaviour under degradation is unknown. The only COD corruption benchmark, COD10K-C [6], evaluates three specialist models on one dataset and includes no SAM-based model. Studies of SAM's robustness [7], [8] use prompted, general-purpose segmentation, not camouflage.

This paper measures how SAM-based camouflage detectors behave under common corruptions and what robustness a student-scale GPU can buy. Our contributions are:

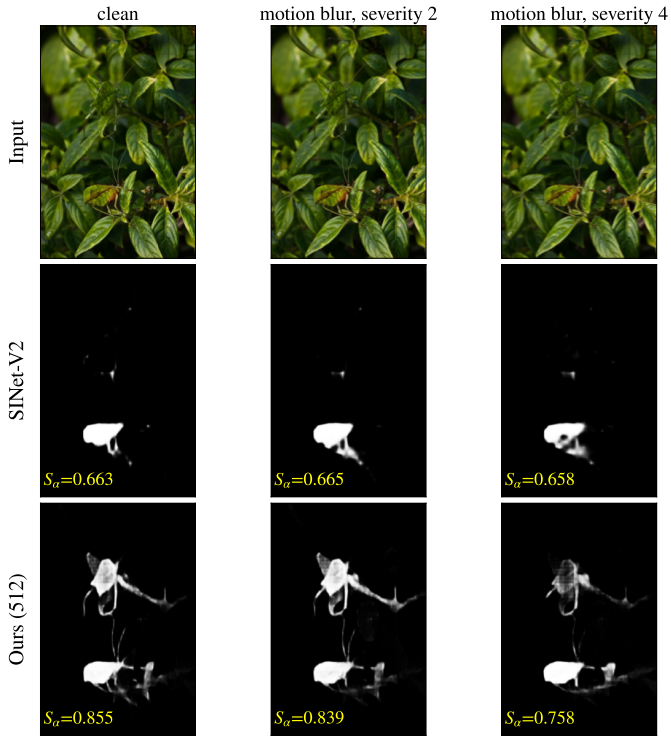


Fig. 1. A COD10K image under increasing motion blur, a corruption never used to train our model. Top: input; middle: SINet-V2; bottom: our adapter at 512 pixels; per-image S_α in each panel. The image is the one at the 90th percentile of our model's gain over SINet-V2 under motion blur (a clear example, not the best one).

- **A benchmark** of three specialist and two SAM-based COD models plus a zero-shot SAM probe, under the eight corruptions and five severities of COD10K-C on four test sets, with paired significance tests.
- **A held-out-corruption protocol**: four corruptions may be used for training, four are only used for testing, so that robustness gains cannot come from memorising the augmentations.
- **An analysis of input resolution**: models that read 1024-pixel inputs lose the most under local corruptions;

training at 512 pixels removes this loss at a small clean-accuracy cost.

- **A laptop-scale robust adapter**: a frozen MobileSAM encoder with 3.71 M trainable parameters, trained in 8.5 h (2.6 h at 512 pixels) on a 4 GB GPU, that outperforms the specialist models under corruption on the two large test sets.

II. Related Work

Camouflaged object detection. COD10K [1] and SINet-V2 [2] established the task and its standard protocol. Specialist models add task-specific modules: distraction mining in PFNet [9], mixed-scale processing in ZoomNet [10] and frequency decomposition in FEDER [11]. CAMO [12], CHAMELEON [13] and NC4K [14] are the other standard test sets.

SAM and adapters for COD. SAM [3] and SAM 2 [15] segment prompted objects but have no notion of camouflage. SAM-Adapter [4] and its successors [16], [17] insert trainable adapters [18], [19] into a frozen SAM encoder; SAM2-UNet [5] pairs a frozen SAM 2 encoder with a U-Net decoder; dual-stream adapters add depth [20]. MobileSAM [21] and EfficientSAM [22] distil SAM into small encoders such as TinyViT [23].

Robustness to corruptions. ImageNet-C [24] introduced corruption benchmarks and the mean corruption error. SAM is robust to most corruptions except blur [7], [8]; Robust-SAM [25] and GaRA-SAM [26] make prompted SAM robust with consistency losses or gated low-rank adapters. For COD, KRNet [27] addresses low-quality inputs and COD10K-C [6] benchmarks specialist models and proposes a lightweight robust model trained with corruption augmentation and a consistency loss. Augmentation with a consistency loss is also the idea behind AugMix [28]. We use these ingredients deliberately and focus on what is new: SAM-family models, unseen corruptions, and the role of input resolution.

III. Benchmark

Datasets. We follow the standard protocol: training on COD10K-train and CAMO-train (4,040 images) and testing on CHAMELEON (76), CAMO (250), COD10K (2,026) and NC4K (4,121). Mean image sizes range from 530×709 (NC4K) to 744×982 (CHAMELEON) pixels.

Corruptions. We use the eight corruptions and five severities of COD10K-C [6]: Gaussian noise ($\sigma = 8$–40), Gaussian blur (radius 0.8–2.5), motion blur (horizontal kernel 3–11), brightness and contrast (factor 0.85–0.30), fog (blend with white, $\alpha = 0.12$–0.44), JPEG (quality 70–20) and rain. COD10K-C does not specify rain numerically; we synthesise streaks of increasing density and length, so rain results are not comparable with theirs. Corruptions are applied to the original image before each model's own resizing, with a seed derived from the dataset, image, corruption and severity, so every model sees identical corrupted images. We call noise, both blurs, JPEG and rain *local* corruptions (they change pixels relative to their neighbours) and brightness, contrast and fog *global* ones.

Held-out protocol. Gaussian noise, Gaussian blur, brightness and JPEG are *seen*: they may be used for training. Motion blur, contrast, fog and rain are *unseen*: each blur and illumination change has a related seen corruption, and weather is held out entirely.

Metrics. We report the structure measure S_α [29], mean E-measure E_ϕ [30], weighted F-measure F_β^w [31], MAE and Dice, computed per image with PySODMetrics [32]. Robustness is summarised by mCorr, the mean score over the 40 corrupted conditions; retention, mCorr divided by the clean score; and mCE, the corruption error relative to SINet-V2 [24]. Differences between models are tested with a paired bootstrap over per-image scores (5,000 resamples, 95% confidence intervals).

Models. We evaluate SINet-V2 [2], PFNet [9], ZoomNet [10], SAM2-UNet [5] and SAM-Adapter (ViT-H) [4] from their released weights, following each official test script, including its input size. FEDER [11] is scored from its released prediction maps on clean images only. SAM2-Adapter and SAM3-Adapter released no COD weights. As an analysis probe, MobileSAM [21] prompted with the ground-truth box shows how SAM delineates an object whose location is known; it is not a competitor. All reproduced clean S_α values are within 0.0014 of the published ones.

IV. Lightweight Robust Adapter

Fig. 2 shows our model. **Encoder.** We freeze the TinyViT encoder of MobileSAM [21], [23]. A bottleneck adapter runs in parallel with the MLP of each of its ten transformer blocks [19]: $x \leftarrow x + \mathrm{MLP}(x) + W_{\mathrm{up}}\,\mathrm{GELU}(W_{\mathrm{down}}\,\mathrm{LN}(x))$, with a hidden width of a quarter of the block width. W_{up} starts at zero, so training begins from exactly the pretrained encoder. **Prompt-free decoding.** Instead of user clicks, the SAM mask decoder receives four learned prompt tokens and a learned dense prompt; two tokens are initialised as SAM's embedding of a box around the whole image. The decoder is trained; its upsampled first mask logit gives the camouflage map.

Training. The supervised loss is the weighted BCE plus weighted IoU structure loss [33]. Each training image gives a clean view and a view corrupted with a random *seen* corruption at a random severity, under the same crop and flip. The loss is

$$\mathcal{L} = \tfrac{1}{2}\big(\mathcal{L}_s(\hat{y}) + \mathcal{L}_s(\hat{y}')\big) + \lambda \left\|\sigma(\hat{y}') - \mathrm{sg}[\sigma(\hat{y})]\right\|_2^2, \quad (1)$$

where $\hat{y}$ and $\hat{y}'$ are the clean and corrupted logits, sg stops gradients, so the clean prediction acts as a fixed target, and $\lambda = 1$. We use AdamW [34] (learning rate 2×10^{-4}, cosine schedule), an effective batch of 8, fp16 and gradient checkpointing, for 30 epochs. *Ours-Clean* uses neither corruption nor consistency; *Ours-Aug* corrupts half of the images without consistency. We train the full model at 1024 pixels, Ours (1024), and at 512 pixels, Ours (512), the latter with three seeds. Training takes 8.5 h and 2.6 h on one RTX 3050 Laptop GPU with 4 GB.

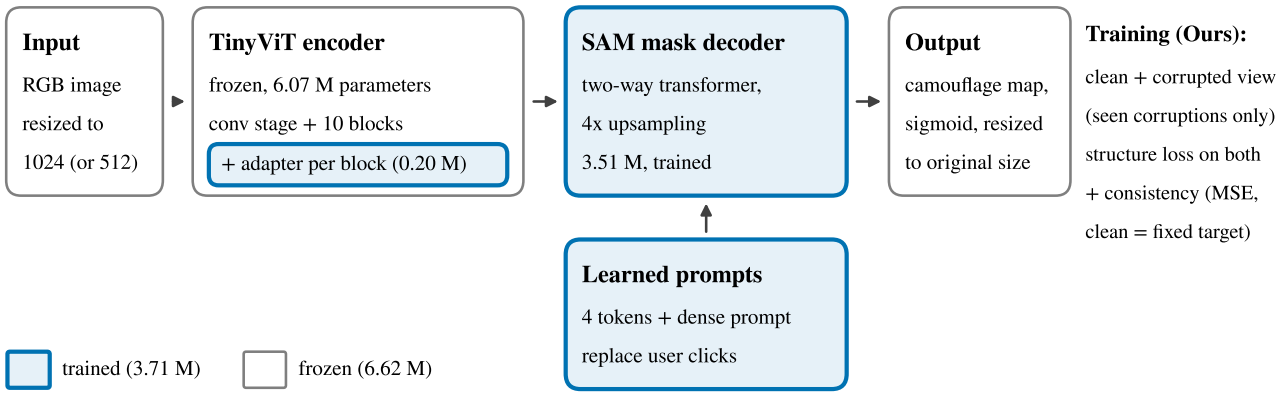


Fig. 2. Our model. Only the coloured parts are trained (3.71 M of 10.33 M parameters). The MobileSAM encoder is frozen; zero-initialised adapters in its ten transformer blocks, four learned prompt tokens and the SAM mask decoder are trained.

V. Experiments

Clean accuracy (Table I). The SAM-based models lead. Ours (1024) reaches 0.840 S_α on COD10K, level with ZoomNet (0.839) and above SINet-V2 (0.815), and Ours (512) reaches 0.823 $\pm$ 0.001 over three seeds. On CHAMELEON, the smallest set, SINet-V2 is ahead of both.

Robustness (Table II, Fig. 3). SAM2-UNet is the most robust model (retention 0.984, mCE 0.66). SAM-Adapter has the best clean score (0.884) but falls behind SAM2-UNet under corruption (0.853 vs. 0.866; difference -0.013, 95% CI [-0.017, -0.009]); on CHAMELEON it drops below SINet-V2 and ZoomNet. Parameter count alone therefore does not predict robustness. Global corruptions cost every model little; rain, blur and noise cause most of the loss: rain lowers Dice retention to 0.828 for ZoomNet and 0.881 for SINet-V2. Under corruption, Ours (1024) outperforms SINet-V2 (+0.017, [+0.013, +0.021]), ZoomNet (+0.009) and PFNet (+0.039) on COD10K, and SINet-V2 by +0.009 on NC4K, all significant (all paired tests are listed in the supplementary material). It ties SINet-V2 on CAMO and trails it on CHAMELEON, and stays below SAM2-UNet (-0.054).

Input resolution (Fig. 4, Table III). The two models that read 1024-pixel inputs keep the least of their score under local corruptions: 0.948 for SAM-Adapter and 0.949 for Ours (1024), against 0.963 for SINet-V2 at 352 pixels. Shrinking a $\approx$740$\times$963 image to 352 pixels averages neighbouring pixels, which weakens noise, JPEG blocks and rain streaks and makes blur smaller, so the same corruption is effectively weaker for a low-resolution model. Shrinking the input of the trained 1024-pixel model only at test time lowers its accuracy everywhere (Table IIIb). Training at 512 pixels instead leaves its accuracy under local corruptions unchanged (-0.000, [-0.003, +0.002]), costs 0.840$\rightarrow$0.823 on clean COD10K, and raises retention to 0.977, the level of SINet-V2. On NC4K and CAMO, whose images are smaller, the 512-pixel model is better than the 1024-pixel one also on clean images (Table I). The useful input size tracks the native image resolution; larger inputs mainly expose the model to corruption.

Held-out corruptions (Table IIIa; per-corruption gains with confidence intervals in the supplementary material). Corruption training improves Ours on seen corruptions by +0.021 S_α on COD10K and on unseen corruptions by +0.005 ([+0.003, +0.007]). The unseen gain comes from motion blur (+0.008) and rain (+0.015), a family absent from training; contrast and fog, which no model is sensitive to, do not change. The consistency loss adds a small further gain over augmentation alone (+0.002, [+0.001, +0.004]). Robustness learned from a few corruptions transfers, but only partly.

Efficiency (Table IV, Fig. 5). Ours (512) runs in 23.1 ms with 87 MB of GPU memory, against 28.6 ms and 134 MB for SINet-V2, while scoring higher than SINet-V2 on COD10K both clean and under corruption (+0.010, [+0.006, +0.013]). Ours (1024) takes 45.6 ms, SAM2-UNet 71.3 ms and SAM-Adapter 3840.6 ms. Both of our models lie on the Pareto front of accuracy under corruption against latency.

Qualitative results (Fig. 6). In the median examples the SAM-based models give the cleanest masks. Our models are comparable to the specialists, with Ours (512) cleaner than Ours (1024), which adds background structures under JPEG compression. The failure cases show the typical breakdown under strong motion blur and rain: the prediction spreads over the whole frame.

VI. Limitations

Our corruptions are synthetic and applied one at a time; real degradations combine and vary across the image, and our rain is our own design. The large models were evaluated but not retrained with corruption augmentation, so robustness training is tested only on our adapter. Two unseen corruptions, contrast and fog, barely affect any model, which limits what the unseen split can show. All timings come from one laptop GPU. Input resolution and architecture differ across the baselines, so our resolution analysis is controlled only for our own model.

VII. Conclusion

SAM-based camouflage detectors are not inherently fragile: SAM2-UNet is both the most accurate and the most robust

TABLE I

CLEAN-IMAGE RESULTS ON THE FOUR COD TEST SETS. BEST PER COLUMN IN BOLD (ORACLE EXCLUDED). †FEDER: SCORED FROM THE AUTHORS' RELEASED PREDICTION MAPS (NO COD10K MAPS RELEASED; TRAINED ON COD10K-TRAIN ONLY). ‡ORACLE: MOBILESAM PROMPTED WITH THE GROUND-TRUTH BOX, AN ANALYSIS PROBE, NOT A COMPETITOR. OURS (512): MEAN ± STD OVER 3 SEEDS.

	CHAMELEON				CAMO				COD10K				NC4K			
Model	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	MAE↓	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	MAE↓	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	MAE↓	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	MAE↓
SINet-V2	0.888	0.941	0.816	0.030	0.820	0.882	0.743	0.070	0.815	0.887	0.680	0.037	0.847	0.902	0.769	0.048
PFNet	0.882	0.931	0.810	0.033	0.782	0.842	0.696	0.085	0.800	0.878	0.660	0.040	0.829	0.887	0.745	0.053
ZoomNet	0.902	0.943	0.844	0.023	0.820	0.877	0.751	0.066	0.839	0.888	0.727	0.029	0.853	0.896	0.784	0.044
FEDER†	0.887	0.946	0.834	0.030	0.802	0.867	0.738	0.071	–	–	–	–	0.847	0.907	0.789	0.044
SAM2-UNet	**0.914**	**0.962**	**0.865**	**0.022**	**0.884**	**0.932**	**0.845**	**0.042**	0.880	**0.937**	0.801	**0.021**	**0.901**	**0.941**	**0.858**	**0.029**
SAM-Adapter	0.896	0.919	0.830	0.032	0.847	0.874	0.774	0.070	**0.884**	0.919	**0.807**	0.025	0.884	0.905	0.818	0.041
Ours (1024)	0.883	0.910	0.777	0.039	0.816	0.834	0.702	0.093	0.840	0.881	0.702	0.040	0.859	0.884	0.755	0.054
Ours (512)	0.851	0.884	0.725	0.045	0.825	0.843	0.710	0.084	$0.823_{\pm.001}$	$0.864_{\pm.001}$	$0.665_{\pm.003}$	$0.042_{\pm.000}$	0.861	0.888	0.758	0.051
MobileSAM + GT box‡	0.787	0.816	0.670	0.109	0.827	0.846	0.746	0.095	0.827	0.864	0.713	0.065	0.850	0.880	0.770	0.072

TABLE II

ROBUSTNESS ON COD10K (8 CORRUPTIONS × 5 SEVERITIES). CLEAN AND MCORR: S_α; RET.: MCORR/CLEAN (S_α UNLESS MARKED); SEEN/UNSEEN: RETENTION ON THE CORRUPTIONS USED / NEVER USED TO TRAIN OUR MODELS; LOCAL: NOISE, BLUR, JPEG, RAIN; GLOBAL: BRIGHTNESS, CONTRAST, FOG; MCE: CORRUPTION ERROR RELATIVE TO SINET-V2. INPUT: SIDE LENGTH IN PX (ZOOMNET* ALSO USES 0.5× AND 1.5×). ‡ORACLE WITH THE GROUND-TRUTH BOX, NOT A COMPETITOR. RETENTION ALONE CAN FAVOUR WEAKER MODELS; READ IT WITH MCORR AND MCE. OURS (512): MEAN ± STD OVER 3 SEEDS.

Model	Input	Clean	mCorr	Ret.	$\text{Ret.}_{\text{Dice}}$	Seen	Unseen	Local	Global	mCE↓
SINet-V2	352	0.815	0.795	0.976	0.956	0.980	0.971	0.963	**0.997**	1.00
PFNet	416	0.800	0.773	0.967	0.937	0.972	0.961	0.958	0.980	1.11
ZoomNet	384*	0.839	0.803	0.957	0.918	0.959	0.956	0.943	0.981	0.96
SAM2-UNet	352	0.880	**0.866**	**0.984**	**0.973**	**0.984**	**0.984**	**0.978**	0.994	**0.66**
SAM-Adapter	1024	**0.884**	0.853	0.964	0.937	0.956	0.973	0.948	0.992	0.72
Ours (1024)	1024	0.840	0.812	0.967	0.946	0.967	0.966	0.949	0.996	0.92
Ours (512)	512	$0.823_{\pm.001}$	$0.804_{\pm.001}$	$0.977_{\pm.000}$	$0.962_{\pm.001}$	$0.978_{\pm.000}$	$0.975_{\pm.001}$	$0.967_{\pm.000}$	$0.993_{\pm.001}$	$0.96_{\pm.00}$
MobileSAM + GT box‡	1024	0.827	0.807	0.976	0.967	0.973	0.978	0.964	0.996	0.94

TABLE III

ABLATIONS. ALL COLUMNS ARE S_α: CLEAN, AND MEAN UNDER CORRUPTION (MCORR) OVER ALL, SEEN, UNSEEN, LOCAL AND GLOBAL CORRUPTIONS. TEST-TIME VIEWS SHRINK THE INPUT OF THE 1024-PX MODEL BEFORE INFERENCE; "TRAINED AT 512 PX" LEARNS AT THAT RESOLUTION. BEST PER BLOCK IN BOLD. OURS (512): MEAN ± STD OVER 3 SEEDS.

Variant	Clean	mCorr	Seen	Unseen	Local	Global
(a) Training components, COD10K						
Adapter, clean images only	**0.842**	0.799	0.791	0.807	0.776	**0.837**
+ corruption augmentation	0.838	0.810	0.809	0.810	0.795	0.834
+ consistency loss (Ours, 1024)	0.840	**0.812**	**0.812**	**0.812**	**0.797**	0.836
Ours, trained at 512 px	$0.823_{\pm.001}$	$0.804_{\pm.001}$	$0.805_{\pm.001}$	$0.803_{\pm.000}$	$0.796_{\pm.001}$	$0.818_{\pm.000}$
(b) Epochs and input resolution, CHAMELEON						
Ours, 15 epochs	0.864	0.827	0.824	0.831	0.810	0.856
Ours, 30 epochs (1024 px)	**0.883**	**0.847**	**0.845**	**0.849**	**0.828**	**0.878**
test-time view 768 px	0.878	0.842	0.842	0.843	0.827	0.868
test-time view 512 px	0.868	0.832	0.831	0.832	0.818	0.854
test-time view 352 px	0.842	0.814	0.816	0.813	0.806	0.829
trained at 512 px	0.851	0.833	0.832	0.833	0.826	0.844

TABLE IV

EFFICIENCY ON ONE NVIDIA GEFORCE RTX 3050 LAPTOP GPU (4 GB), BATCH 1, END-TO-END ON A 768×1024 IMAGE (MEDIAN OVER RUNS). PRECISION AS IN EACH MODEL'S OFFICIAL INFERENCE. TRAINING: OUR FULL TRAINING RUN ON THE SAME LAPTOP GPU.

Model	Input	Prec.	Params (M)	GFLOPs	Latency (ms)	FPS	VRAM (MB)	Train (h)
SINet-V2	352	fp32	27.0	24.3	28.6	35.0	134	–
PFNet	416	fp32	46.5	53.0	37.8	26.4	258	–
ZoomNet	384*	fp32	32.4	202.7	94.4	10.6	410	–
SAM2-UNet	352	fp16	216.5	278.8	71.3	14.0	879	–
SAM-Adapter	1024	fp16	641.3	5979.2	3840.6	0.3	2477	–
Ours (1024)	1024	fp16	**10.3**	83.2	45.6	21.9	181	8.5
Ours (512)	512	fp16	**10.3**	**22.1**	**23.1**	**43.2**	**87**	2.6

model we tested. Size alone does not buy robustness, and local corruptions, especially rain, blur and noise, do almost all the damage. A model's input resolution decides how strongly these corruptions reach it. A frozen MobileSAM with 3.71 M trainable parameters, trained on a 4 GB laptop GPU with corruption augmentation, outperforms the specialist models under corruption on the large test sets, and at 512 pixels it matches their robustness at lower cost. Future work includes resolution-aware training, multi-scale decoder features [35] and real-world degraded test data.

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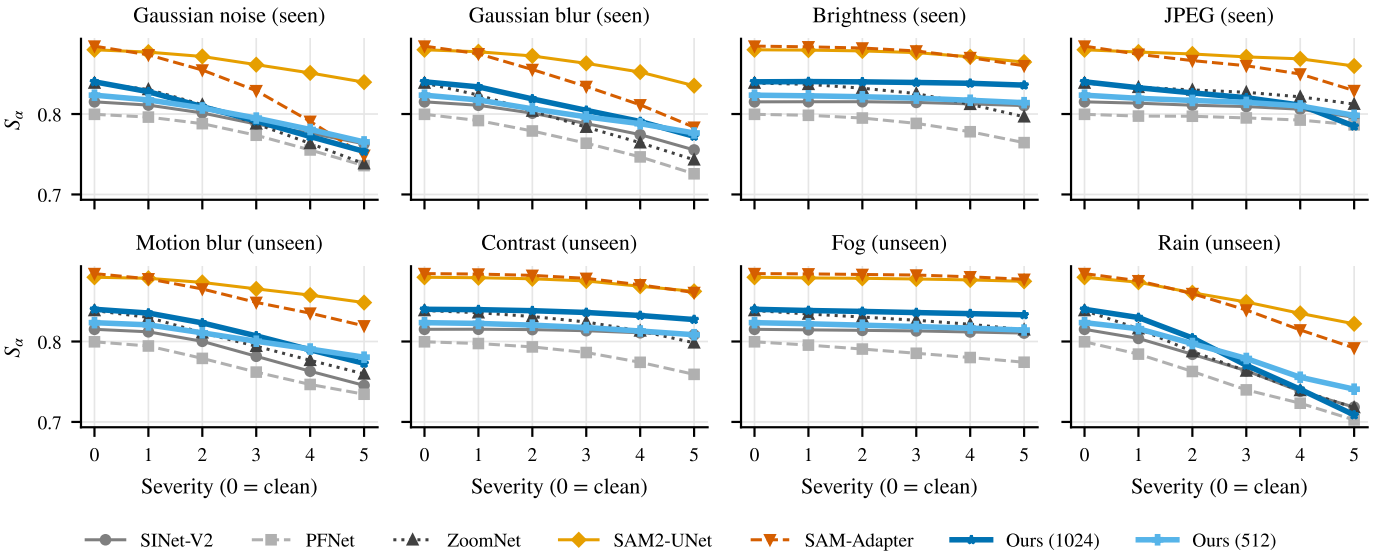


Fig. 3. S_α versus severity for each corruption on COD10K (severity 0 is clean). The top row shows corruptions used to train our models, the bottom row unseen ones.

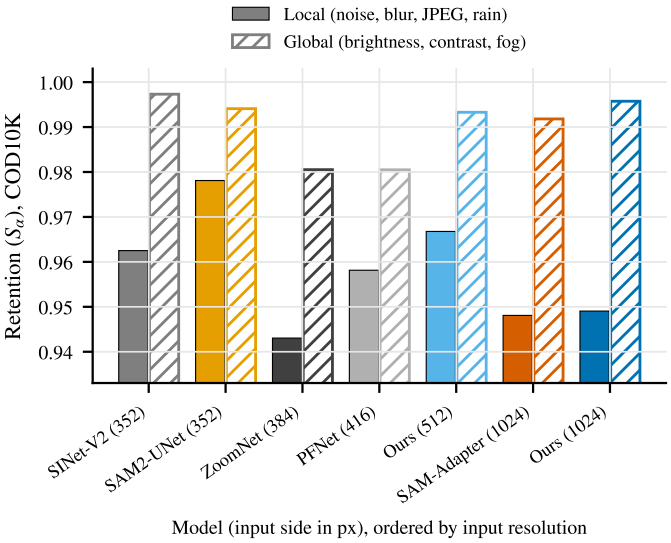


Fig. 4. Retention under local and global corruptions on COD10K, with models ordered by input side length. The two 1024-pixel models keep the least under local corruptions.

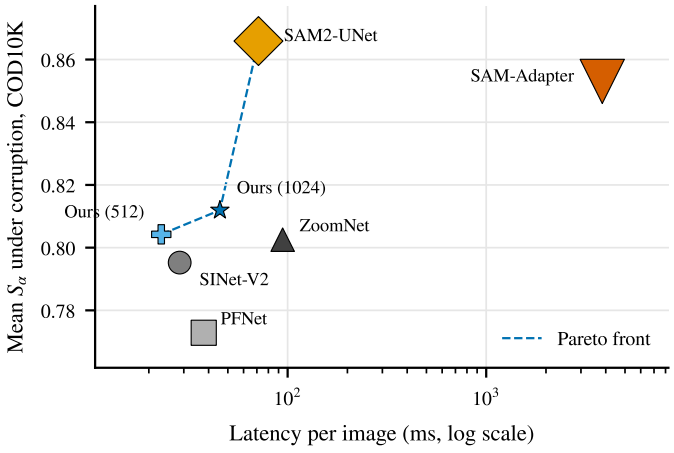


Fig. 5. Accuracy under corruption against latency on COD10K (RTX 3050 Laptop GPU). Marker area grows with parameter count. Our two models lie on the Pareto front between SINet-V2 and SAM2-UNet.

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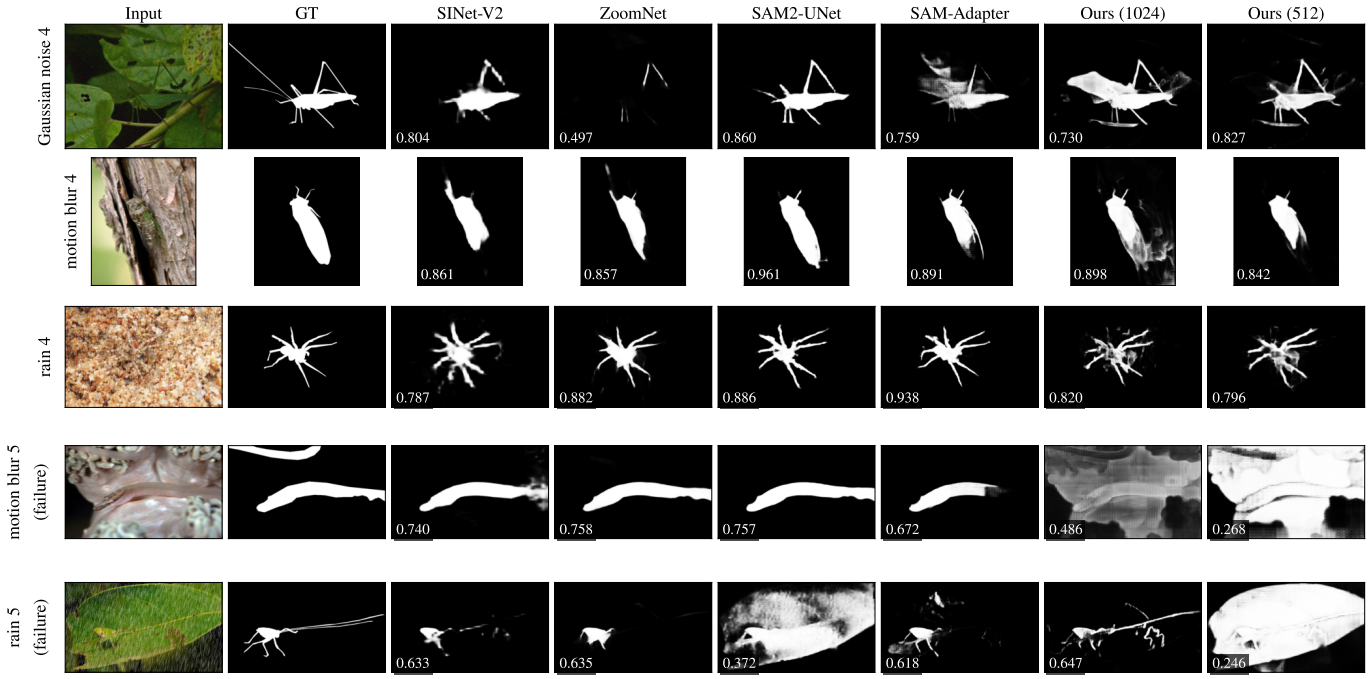


Fig. 6. Predictions on COD10K. The first three rows show, for each corruption, the image on which Ours (512) has its median score; the last two rows show its lowest-scoring images (failure cases). Numbers are per-image S_α. More rows are in the supplementary material.

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